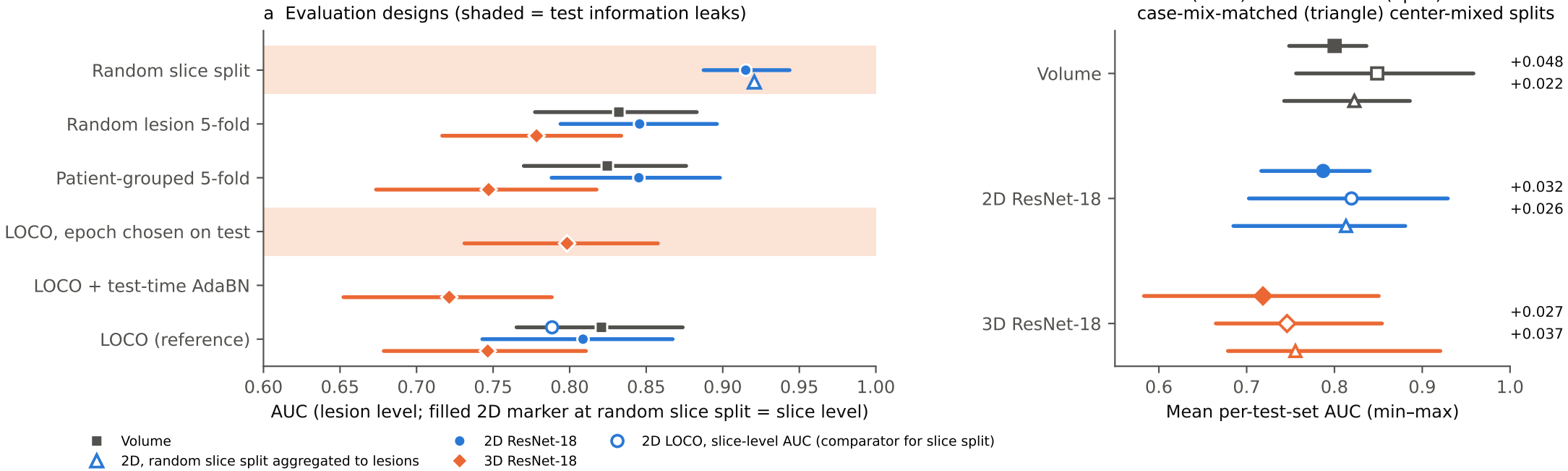
a Evaluation designs (shaded = test information leaks)
Random slice split
Random lesion 5-fold
Patient-grouped 5-fold
LOCO, epoch chosen on test
LOCO + test-time AdaBN
LOCO (reference)
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95
1.00
AUC (lesion level; filled 2D marker at random slice split = slice level)
b LOCO (filled) · size-matched (open) ·
case-mix-matched (triangle) center-mixed splits
Volume
2D ResNet-18
3D ResNet-18
+0.048
+0.022
+0.032
+0.026
+0.027
+0.037
0.6
0.7
0.8
0.9
1.0
Mean per-test-set AUC (min–max)
Volume
2D, random slice split aggregated to lesions
2D ResNet-18
3D ResNet-18
2D LOCO, slice-level AUC (comparator for slice split)